Powers of 10

Name _____ **Date** _____ No calculator Version 1

KEY WORDS power · exponent · power of 10 · tenth · hundredth · thousandth · expanded form

A Positive powers of 10 ●○○

ANSWER ON THE SHEET

Remember: The power tells you how many zeros follow the 1: $10^3 = 10 \times 10 \times 10 = 1{,}000$.

1. Write each power of 10 as an ordinary number.

a) 10^2 = _____ **b)** 10^3 = _____ **c)** 10^1 = _____ **d)** 10^5 = _____

e) 10^4 = _____ **f)** 10^0 = _____ **g)** 10^6 = _____ **h)** 10^8 = _____

2. Write each number as a power of 10.

a) 1,000 = _____ **b)** 100 = _____ **c)** 100,000 = _____

d) 10 = _____ **e)** 1,000,000 = _____ **f)** 10,000 = _____

g) 1 = _____ **h)** 1,000,000,000 = _____ **i)** 10,000,000 = _____

B Negative powers of 10 ●●○

ANSWER ON THE SHEET

Watch out: A negative power does not make the number negative. It means "one over": $10^{-2} = \frac{1}{10^2} = \frac{1}{100} = 0.01$.

3. Write each power of 10 as a decimal.

a) 10^{-1} = _____ **b)** 10^{-2} = _____ **c)** 10^{-3} = _____ **d)** 10^{-4} = _____

4. Write each power of 10 as a fraction.

a) 10^{-1} = _____ **b)** 10^{-3} = _____ **c)** 10^{-2} = _____ **d)** 10^{-5} = _____

5. Write each number as a power of 10.

a) 0.01 = _____ **b)** $\frac{1}{10}$ = _____ **c)** 0.001 = _____

d) $\frac{1}{10{,}000}$ = _____ **e)** 0.1 = _____ **f)** $\frac{1}{100}$ = _____

g) 0.00001 = _____ **h)** $\frac{1}{1{,}000{,}000}$ = _____ **i)** 0.0001 = _____

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C Multiplying by powers of 10 ●●○ — ANSWER ON THE SHEET

> **Remember:** Multiplying by 10^2 moves every digit 2 places to the left. Multiplying by 10^{-2} moves every digit 2 places to the right.

6. Work out.

a) 4.2×100 = _____ **b)** 0.37×10 = _____ **c)** $15 \times 1{,}000$ = _____ **d)** 0.56×10^3 = _____

e) 2.08×10^2 = _____ **f)** 0.009×10^2 = _____ **g)** 4.05×10^4 = _____ **h)** 0.6×10^1 = _____

i) 3.4×10^{-1} = _____ **j)** 6×10^{-2} = _____ **k)** 125×10^{-3} = _____ **l)** 70×10^{-1} = _____

m) 9.2×10^{-2} = _____ **n)** 480×10^{-2} = _____ **o)** 5×10^{-3} = _____ **p)** 36×10^0 = _____

D Dividing by powers of 10 ●●○ — ANSWER ON THE SHEET

> **Remember:** Dividing by 10^2 moves every digit 2 places to the right, so the number gets smaller.

7. Work out.

a) $350 \div 10$ = _____ **b)** $78 \div 100$ = _____ **c)** $6.4 \div 10$ = _____ **d)** $29 \div 1{,}000$ = _____

e) $78 \div 10^2$ = _____ **f)** $0.9 \div 10^2$ = _____ **g)** $45 \div 10^3$ = _____ **h)** $8 \div 10^1$ = _____

i) $1{,}260 \div 10^2$ = _____ **j)** $0.3 \div 10^1$ = _____ **k)** $507 \div 10^3$ = _____ **l)** $600 \div 10^4$ = _____

E Expanded form ●●● — WORK IN YOUR BOOK

> **Remember:** Each digit is multiplied by the power of 10 for its place: $3.61 = 3 \times 10^0 + 6 \times 10^{-1} + 1 \times 10^{-2}$.

8. Write as an ordinary number.

a) $4 \times 10^0 + 7 \times 10^{-1}$

b) $2 \times 10^1 + 5 \times 10^0 + 3 \times 10^{-1}$

c) $6 \times 10^{-1} + 2 \times 10^{-2}$

d) $8 \times 10^0 + 4 \times 10^{-2}$

e) $3 \times 10^2 + 9 \times 10^{-1}$

f) $1 \times 10^2 + 3 \times 10^{-1} + 5 \times 10^{-3}$

9. Write each number in expanded form, using powers of 10.

a) 52

b) 3.61

c) 0.48

d) 40.06

e) 0.507

f) 7.285

★ Extension — FOR EARLY FINISHERS

10. Find n if $10^n = 0.00001$.

11. How many times larger is 10^3 than 10^{-2}?

12. Work out $2.5 \div 10^{-1}$. (Think: dividing by a tenth is the same as multiplying by what?)

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Powers of 10

Name _____ **Date** _____ No calculator Version 2

KEY WORDS power · exponent · power of 10 · tenth · hundredth · thousandth · expanded form

A Positive powers of 10 ●○○ ANSWER ON THE SHEET

Remember: The power tells you how many zeros follow the 1: $10^3 = 10 \times 10 \times 10 = 1{,}000$.

1. Write each power of 10 as an ordinary number.

a) 10^3 = _____ **b)** 10^2 = _____ **c)** 10^4 = _____ **d)** 10^1 = _____

e) 10^6 = _____ **f)** 10^0 = _____ **g)** 10^5 = _____ **h)** 10^7 = _____

2. Write each number as a power of 10.

a) 100 = _____ **b)** 10,000 = _____ **c)** 1,000 = _____

d) 10 = _____ **e)** 100,000 = _____ **f)** 10,000,000 = _____

g) 1 = _____ **h)** 100,000,000 = _____ **i)** 1,000,000 = _____

B Negative powers of 10 ●●○ ANSWER ON THE SHEET

⚠ **Watch out:** A negative power does not make the number negative. It means "one over": $10^{-2} = \frac{1}{10^2} = \frac{1}{100} = 0.01$.

3. Write each power of 10 as a decimal.

a) 10^{-2} = _____ **b)** 10^{-1} = _____ **c)** 10^{-4} = _____ **d)** 10^{-3} = _____

4. Write each power of 10 as a fraction.

a) 10^{-2} = _____ **b)** 10^{-1} = _____ **c)** 10^{-4} = _____ **d)** 10^{-3} = _____

5. Write each number as a power of 10.

a) 0.1 = _____ **b)** $\frac{1}{100}$ = _____ **c)** 0.0001 = _____

d) $\frac{1}{1{,}000}$ = _____ **e)** 0.01 = _____ **f)** $\frac{1}{10}$ = _____

g) 0.000001 = _____ **h)** $\frac{1}{100{,}000}$ = _____ **i)** 0.001 = _____

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C Multiplying by powers of 10 ●●○ — ANSWER ON THE SHEET

> **Remember:** Multiplying by 10^2 moves every digit 2 places to the left. Multiplying by 10^{-2} moves every digit 2 places to the right.

6. Work out.

a) 5.3×100 = ____ **b)** 0.48×10 = ____ **c)** $27 \times 1{,}000$ = ____ **d)** 0.72×10^3 = ____

e) 6.04×10^2 = ____ **f)** 0.007×10^2 = ____ **g)** 3.06×10^4 = ____ **h)** 0.9×10^1 = ____

i) 5.8×10^{-1} = ____ **j)** 4×10^{-2} = ____ **k)** 375×10^{-3} = ____ **l)** 90×10^{-1} = ____

m) 7.4×10^{-2} = ____ **n)** 650×10^{-2} = ____ **o)** 8×10^{-3} = ____ **p)** 52×10^0 = ____

D Dividing by powers of 10 ●●○ — ANSWER ON THE SHEET

> **Remember:** Dividing by 10^2 moves every digit 2 places to the right, so the number gets smaller.

7. Work out.

a) $470 \div 10$ = ____ **b)** $63 \div 100$ = ____ **c)** $8.2 \div 10$ = ____ **d)** $37 \div 1{,}000$ = ____

e) $94 \div 10^2$ = ____ **f)** $0.5 \div 10^2$ = ____ **g)** $28 \div 10^3$ = ____ **h)** $6 \div 10^1$ = ____

i) $2{,}340 \div 10^2$ = ____ **j)** $0.7 \div 10^1$ = ____ **k)** $409 \div 10^3$ = ____ **l)** $900 \div 10^4$ = ____

E Expanded form ●●● — WORK IN YOUR BOOK

> **Remember:** Each digit is multiplied by the power of 10 for its place: $3.61 = 3 \times 10^0 + 6 \times 10^{-1} + 1 \times 10^{-2}$.

8. Write as an ordinary number.

a) $6 \times 10^0 + 3 \times 10^{-1}$

b) $4 \times 10^1 + 7 \times 10^0 + 2 \times 10^{-1}$

c) $8 \times 10^{-1} + 5 \times 10^{-2}$

d) $5 \times 10^0 + 7 \times 10^{-2}$

e) $5 \times 10^2 + 4 \times 10^{-1}$

f) $2 \times 10^2 + 1 \times 10^{-1} + 6 \times 10^{-3}$

9. Write each number in expanded form, using powers of 10.

a) 84

b) 2.95

c) 0.73

d) 60.02

e) 0.309

f) 4.618

★ Extension — FOR EARLY FINISHERS

10. Find n if $10^n = 0.000001$.

11. How many times larger is 10^2 than 10^{-3}?

12. Work out $3.6 \div 10^{-2}$. (Think: dividing by a tenth is the same as multiplying by what?)

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